



Seunghoon Woo

Assistant Professor (@KOREA UNIVERSITY), Chief Scientist (@LABRADOR LABS)
seunghoonwoo@korea.ac.kr | <https://ssp.korea.ac.kr>

SOFTWARE SECURITY; SOFTWARE VULNERABILITY DETECTION;
SOFTWARE COMPOSITION ANALYSIS; CODE CLONE DETECTION;
SECURITY FOR AI; AI FOR SECURITY.

EARNED DEGREES

- **M.S. & Ph.D.** in Computer Science and Engineering, Korea University (GPA 4.45/4.5) Sep 2016 - Aug 2022
- **B.S.** in Computer Science and Engineering, Korea University (GPA 4.22/4.5) Mar 2010 - Feb 2016

DOCTORAL DISSERTATION

- Detecting Software Vulnerabilities for Mitigating Risks of Open-Source Reuse (Advisor: Prof. Heejo Lee) Aug 2022

WORKING EXPERIENCES

- Korea University, **Assistant Professor** Sep 2023 - Present
- LABRADOR LABS Inc., **Chief Scientist** May 2022 - Present
- Center for Software Security and Assurance (CSSA), Research Professor Sep 2022 - Aug 2023
- National University of Singapore, Research Intern Dec 2016 - Feb 2017
- Samsung Electronics, Student Intern & Employee Jun 2014 - Aug 2014, Dec 2015 - Jan 2016
- DoDotDo (startup), Core Developer Jan 2015 - Sep 2015

COMMITTEE

- USENIX Security 2027 (Security 2027) 2027
- IEEE/ACM International Conference on Automated Software Engineering 2026 (ASE 2026) 2026
- USENIX Security 2026 (Security 2026) 2026
- Information Security Conference (ISC 2025) 2025
- ACM ASIA Conference on Computer and Communications Security (ASIACCS 2025) 2025

REVIEWER EXPERIENCES

- TBD: Transactions on Big Data 2026
- COSE: Computers & Security 2025, 2026
- T-IFS: IEEE Transactions on Information Forensics and Security 2025
- TDSC: IEEE Transactions on Dependable and Secure Computing 2024, 2026
- TSE: IEEE Transactions on Software Engineering 2023, 2024
- TOSEM: ACM Transactions on Software Engineering and Methodology 2023
- SP&E: Software: Practice and Experience 2023
- IEEE Transactions on Vehicular Technology 2022
- Journal of Communications and Networks 2021

- [1] **GANADI: Uncovering C/C++ OSS Reuse Genealogies via Pivotal Function-based Clustering to Enhance Supply Chain Security**
Dongyeon Kim, Seunghoon Woo*, and Heejo Lee* (* Co-corresponding authors)
ASE 2026: 41st IEEE/ACM Automated Software Engineering Conference (**Top-tier**)
Munich, Germany, Oct 2026
- [2] **Uncovering Similar but Different Packages in PyPI and Potential Security Threats**
Sunha Park, Soojin Han, Seunghoon Woo
FSE 2026: The ACM International Conference on the Foundations of Software Engineering (**Top-tier**)
Montreal, Canada, Jul 2026
- [3] **SBridge: Identifying Source-to-Binary Function Similarity via Cross-Domain Control Block Matching**
Heedong Yang, Jeongwoo Lee, Hajin Yun, Seunghoon Woo
FSE 2026: The ACM International Conference on the Foundations of Software Engineering (**Top-tier**)
Montreal, Canada, Jul 2026
- [4] **Cloverly: Identifying Affected Versions in C/C++ Public Security Vulnerability Reports**
Duyeong Kim, Jimin Kang, Yeonhee Kim, Seunghoon Woo, and Heejo Lee
IFIPSEC 2026: 41st IFIP international conference on ICT systems security and privacy protection
Perth, Australia, Jun 2026
- [5] **Human vs AI: Insecure Use of Security-Sensitive Functions in AI-generated Code in Comparison to Human-written Code**
Joonku Lee, Sieun Ju, Seunghoon Woo, and Heejo Lee
ACM SAC 2026: 41st ACM/SIGAPP Symposium On Applied Computing
Thessaloniki, Greece, Mar 2026
- [6] **CRYPTBARA: Dependency-Guided Detection of Python Cryptographic API Misuses**
Seogyong Cho, Seungeun Yu, and Seunghoon Woo
ASE 2025: 40th IEEE/ACM Automated Software Engineering Conference (**Top-tier**)
Seoul, Korea, Nov 2025
- [7] **ZCOVER: Uncovering Z-Wave Controller Vulnerabilities Through Systematic Security Analysis of Application Layer Implementation**
Carlos Nkuba Kayembe, Jimin Kang, Seunghoon Woo*, and Heejo Lee* (* Co-corresponding authors)
DSN 2025: 55th Annual IEEE/IFIP International Conference on Dependable Systems and Networks
Naples, Italy, Jun 2025
- [8] **TIVER: Identifying Adaptive Versions of C/C++ Third-Party Open-Source Components Using a Code Clustering Technique**
Youngjae Choi and Seunghoon Woo
ICSE 2025: 47th International Conference on Software Engineering (**Top-tier**)
Ottawa, Canada, Apr 2025
- [9] **Enhancing Code Vulnerability Detection Using CodeGraphBERT Deep Learning Techniques**
Zeinab Shahbazi, Meshkat Mesbah, and Seunghoon Woo
ICIAI 2025: 9th International Conference on Innovation in Artificial Intelligence
Singapore, Singapore, Mar 2025
- [10] **BLOOMFUZZ: Unveiling Bluetooth L2CAP Vulnerabilities via State Cluster Fuzzing with Target-Oriented State Machines**

Pyeongju Ahn, Yeonseok Jang, Seunghoon Woo*, and Heejo Lee* (* Co-corresponding authors)
29th European Symposium on Research in Computer Security (ESORICS 2024)
Bydgoszcz, Poland, Sep 2024

- [11] **CNEPS: A Precise Approach for Examining Dependencies among Third-Party C/C++ Open-Source Components**
Yoonjong Na, Seunghoon Woo*, Joomyeong Lee, and Heejo Lee* (* Co-corresponding authors)
ICSE 2024: International Conference on Software Engineering (**Top-tier**)
Lisbon, Portugal, Apr 2024
- [12] **V1SCAN: Discovering 1-day Vulnerabilities in Reused C/C++ Open-source Software Components Using Code Classification Techniques**
Seunghoon Woo, Eunjin Choi, Heejo Lee, and Hakjoo Oh
Security 2023: 32nd USENIX Security Symposium (**Top-tier**)
Anaheim, USA, Aug 2023
- [13] **MOVERY: A Precise Approach for Modified Vulnerable Code Clone Discovery from Modified Open-Source Software Components**
Seunghoon Woo, Hyunji Hong, Eunjin Choi, and Heejo Lee
Security 2022: 31st USENIX Security Symposium (**Top-tier**)
Boston, USA, Aug 2022
- [14] **L2Fuzz: Discovering Bluetooth L2CAP Vulnerabilities Using Stateful Fuzz Testing**
Haram Park, Carlos Nkuba Kayembe, Seunghoon Woo, and Heejo Lee
DSN 2022: 52nd IEEE/IFIP International Conference on Dependable Systems and Networks
Baltimore, USA, Jun 2022
- [15] **DICOS: Discovering Insecure Code Snippets from Stack Overflow Posts by Leveraging User Discussions**
Hyunji Hong, Seunghoon Woo, and Heejo Lee
ACSAC 2021: Annual Computer Security Applications Conference
Virtual, Dec 2021
- [16] **V0Finder: Discovering the Correct Origin of Publicly Reported Software Vulnerabilities**
Seunghoon Woo, Dongwook Lee, Sunghan Park, Heejo Lee, and Sven Dietrich
Security 2021: 30th USENIX Security Symposium (**Top-tier**)
Virtual, Aug 2021
- [17] **OctoPoCs: Automatic Verification of Propagated Vulnerable Code Using Reformed Proofs of Concept**
Seongkyeong Kwon, Seunghoon Woo, Gangmo Seong, and Heejo Lee
DSN 2021: 51st IEEE/IFIP International Conference on Dependable Systems and Networks
Virtual, Jun 2021
- [18] **CENTRIS: A Precise and Scalable Approach for Identifying Modified Open-Source Software Reuse**
Seunghoon Woo, Sunghan Park, Seulbae Kim, Heejo Lee, and Hakjoo Oh
ICSE 2021: 43rd International Conference on Software Engineering (**Top-tier**)
Virtual, May 2021
- [19] **UDDY: A Scalable Approach for Vulnerable Code Clone Discovery**
Seulbae Kim, Seunghoon Woo, Heejo Lee, and Hakjoo Oh
S&P 2017: 38th IEEE Symposium on Security and Privacy (**Top-tier**)
San Jose, USA, May 2017

- [1] **A Large-Scale Analysis of the Effectiveness of Publicly Reported Security Patches**
Seunghoon Woo, Eunjin Choi, and Heejo Lee
Computers & Security (SCIE/IF: 4.8), Jan 2025
- [2] **ZMAD: Lightweight Model-based Anomaly Detection for the Structured Z-Wave Protocol**
Carlos Nkuba Kayembe, Seunghoon Woo, Heejo Lee, Sven Dietrich
IEEE ACCESS (SCIE/IF: 3.476), Jun 2023
- [3] **CIRCUIT: A JavaScript Memory Heap-Based Approach for Precisely Detecting Cryptojacking Websites**
Seunghoon Woo*, Hyunji Hong*, Sunghan Park*, Jeongwook Lee, and Heejo Lee (* contributed equally)
IEEE ACCESS (SCIE/IF: 3.476), Sep 2022
- [4] **xVDB: A High-Coverage Approach for Constructing a Vulnerability Database**
Hyunji Hong, Seunghoon Woo, Eunjin Choi, Jihyun Choi, and Heejo Lee
IEEE ACCESS (SCIE/IF: 3.476), Aug 2022

PUBLICATIONS - DOMESTIC

- [1] **AI 에이전트의 실행 수준 취약점 탐지를 위한 Attack Chain 기반 Taint 분석**
도현정, 이한, 조서경, 우승훈, *한국컴퓨터종합학술대회*, 2026
- [2] **LLM 코드리뷰에서 PR 메타데이터를 통한 간접 프롬프트 주입 공격의 실증 분석**
한호진, 박선하, 양희동, 우승훈, *한국컴퓨터종합학술대회*, 2026
- [3] **Cross-Language 경로 분석 기반 npm 네이티브 익스텐션 취약점 탐지**
홍민혁, 양희동, 최영재, 우승훈, *한국컴퓨터종합학술대회*, 2026
- [4] **Java 생태계에서 SCA 분석 기법의 한계와 연구 동향**
양소은, 양희동, 유승은, 우승훈, *한국컴퓨터종합학술대회*, 2026
- [5] **LLM 기반 Vulnerability Patch Localization 기술 연구**
금도혁, 우승훈, *한국컴퓨터종합학술대회*, 2026
- [6] **COTS 바이너리의 1-day 취약점 탐지를 위한 Patch Presence Test 연구**
양희동, 이정우, 우승훈, *한국정보보호학회 CISC-S*, 2025
- [7] **공급망 보안을 위한 소프트웨어 명세서(SBOM) 개선 연구**
Youngjae Choi, Heedong Yang, and Seunghoon Woo, *정보보호학회지*, 2025
- [8] **Web 3.0 시대 핵심 기술, 블록체인 보안 위협 전망 및 분석**
Seunghoon Woo, Geonwoo Lee, Taejun Lee, Yunseong Choi, Heejo Lee, Kyeongsik Min, and Jinsang Park, *KISA INSIGHT*, 2023
- [9] **오픈소스 SW 취약점 분석 및 탐지기술 동향**
Seunghoon Woo, Hyunji Hong, and Heejo Lee, *OSIA Standards & Technology Review*, 2022
- [10] **공급망 보안을 위한 오픈소스 소프트웨어 취약점 관리 기술**
Hyunji Hong, Seunghoon Woo, and Heejo Lee, *Review of KIISC*, 2022

PATENT

- [1] **METHOD AND APPARATUS FOR DETECTING PROPAGATION OF SECURITY VULNERABILITIES OF OPEN SOURCE SOFTWARE INHERENT IN COMPONENTS OF TARGET SOFTWARE**
Heejo Lee and Seunghoon Woo
REGISTRATION, US (12,694,126 B2), Jul 2026

- [2] **METHOD AND APPARATUS FOR DETECTING PROPAGATION OF SECURITY VULNERABILITIES OF OPEN SOURCE SOFTWARE INHERENT IN COMPONENTS OF TARGET SOFTWARE**
Heejo Lee and Seunghoon Woo
REGISTRATION, Korea (10-2977203), Jul 2026
- [3] **METHOD AND APPARATUS FOR DETECTING MISUSE OF APPLICATION PROGRAMMING INTERFACE IN PYTHON SOURCE CODE**
Seunghoon Woo and Seogyong Cho
APPLICATION, Korea (10-2026-0075459), Apr 2026
- [4] **METHOD AND APPARATUS FOR GENERATING ADAPTIVE VERSION COVERING VERSION DISTRIBUTION OF REUSED OPEN SOURCE SOFTWARE COMPONENT**
Seunghoon Woo and Youngjae Choi
APPLICATION, Korea (10-2025-0123459), Sep 2025
- [5] **METHOD FOR IDENTIFYING OPEN-SOURCE SOFTWARE COMPONENTS AT THE SOURCE-CODE LEVEL**
Heejo Lee and Seunghoon Woo
REGISTRATION, Europe (EP4033380B1), Apr 2025
- [6] **METHOD FOR IDENTIFYING OPEN-SOURCE SOFTWARE COMPONENTS AT THE SOURCE-CODE LEVEL**
Heejo Lee and Seunghoon Woo
REGISTRATION, US (11836486), Dec 2023
- [7] **METHOD FOR IDENTIFYING OPEN-SOURCE SOFTWARE COMPONENTS AT THE SOURCE-CODE LEVEL**
Heejo Lee and Seunghoon Woo
REGISTRATION, Korea (10-2476358), Dec 2022

STANDARD

- **Structured Software Vulnerability Database Information Expression for Vulnerability Detection and Resolution**
Heejo Lee, Seunghoon Woo, Hyunji Hong, Choonsik Park, and Yunseong Choi
Korea (TTAK.KO-12.0384), Jun 2022

PROJECTS (SELECTED)

- **Research on Intelligent SBOM Generation and Automated Vulnerability Analysis through Multi-level Code Analysis**
Principal Investigator, Mar 2025 - Present
- **Development of Automated SBOM and VEX Verification Technologies for Securing Software Supply Chains**
Co-Investigator, Jul 2024 - Present
- **International Collaborative Research and Global Talent Development for the Development of Copyright Management and Protection Technologies for Generative AI**
Co-Investigator, Apr 2024 - Present
- **Development of SBOM Technologies for Securing Software Supply Chains (IITP/MSIT)**
Researcher, Apr 2022 - Dec 2025
- **Development of Automated Vulnerability Discovery Technologies for Blockchain Security (IITP/MSIT)**
Project Manager & Researcher & Developer, International Joint Research (ETH Zurich), Jun 2019 - Dec 2022

- **The Intelligent IoT Integrator (I3): LA Smart City Project**
Researcher, International Joint Research (City of LA, University of Southern California, etc.), Nov 2017 - Present
- **Verifying Open-Source Software Reliability for Reinforcing Operating System Security (NSR)**
Researcher & Developer, Apr 2020 - Oct 2020
- **Examining Software Vulnerabilities on Platform for IoT-based Home Appliance Consulting Service (KETI)**
Researcher & Analyst, Feb 2020 - Apr 2020
- **Development of DNS-based Lightweight Framework for Addressing Abnormal Network Behaviors (KISTI)**
Researcher & Developer, May 2018 - Oct 2018
- **A Study of a DDoS-resilient Network Architecture through Traffic Classification and Isolation (US ONR)**
Project Manager, International Joint Research (ETH Zurich, Office of Naval Research), Sep 2017 - Sep 2019
- **Development of Vulnerability Discovery Technologies for IoT Software Security (IITP/MSIT)**
Researcher, International Joint Research (ETH Zurich, CMU, University of Oxford), Feb 2016 - May 2018

OPEN-SOURCE SOFTWARE ARTIFACTS

- [1] **SBRIDGE**, A tool for identifying binary functions similar to an input source function (FSE 2026)
https://github.com/heedongy/SBridge_Artifact
- [2] **CRYPTBARA**, A tool for identifying cryptographic API miuses in Python (ASE 2025)
<https://github.com/choseogyong/CRYPTBARA>
- [3] **TIVER**, A tool for identifying adaptive versions of OSS components (ICSE 2025)
<https://github.com/Genius-Choi/TIVER-public>
- [4] **CNEPS**, A tool for discovering OSS component dependencies (ICSE 2024)
<https://github.com/sodium49/CNEPS-public>
- [5] **V1SCAN**, A tool for discovering 1-day security vulnerabilities (Security 2023)
<https://github.com/WOOSEUNGHOOON/V1SCAN-public>
- [6] **MOVERY**, A tool for discovering propagated vulnerable codes (Security 2022)
<https://github.com/WOOSEUNGHOOON/MOVERY-public>
- [7] **VOFinder**, A tool for discovering the correct origin of software vulnerabilities (Security 2021)
<https://github.com/WOOSEUNGHOOON/VOFinder-public>
- [8] **CENTRIS**, A tool for identifying open-source software components (ICSE 2021)
<https://github.com/WOOSEUNGHOOON/CENTRIS-public>